Appendix B

Automating Injury Surveillance: LLMs for Clinical Data Extraction

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Accuracy Tables

Cloud Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.5197	0.0282
Average Field Accuracy	0.9209	0.0057

Table 1: GPT-5: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8894	0.0126
Industry	0.9510	0.0038
Severity	0.7787	0.0181
Mode of Transport	0.9729	0.0072
Cause of Injury	0.9990	0.0015
Injury Mechanism	0.8655	0.0106
Injury Location	0.9107	0.0043
Time of Injury	1.0000	0.0000

Table 2: GPT-5: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3363	0.0185
Avg Field Accuracy	0.8789	0.0047

Table 3: GPT-4o: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8105	0.0160
Industry	0.9247	0.0046
Severity	0.7527	0.0149
Mode of Transport	0.9390	0.0082
Cause of Injury	0.9476	0.0154
Injury Mechanism	0.8384	0.0117
Injury Location	0.8207	0.0116
Time of Injury	0.9973	0.0026

Table 4: GPT-4o: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4797	0.0113
Avg Field Accuracy	0.9125	0.0026

Table 5: Gemini flash 2.5: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9065	0.0052
Industry	0.9697	0.0033
Severity	0.7348	0.0195
Mode of Transport	0.9416	0.0044
Cause of Injury	0.9961	0.0013
Injury Mechanism	0.8484	0.0092
Injury Location	0.9081	0.0106
Time of Injury	0.9952	0.0033

Table 6: Gemini flash 2.5: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.5135	0.0079
Avg Field Accuracy	0.9158	0.0017

Table 7: Grok-3: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9155	0.0053
Industry	0.9513	0.0035
Severity	0.7628	0.0065
Mode of Transport	0.9642	0.0040
Cause of Injury	0.9938	0.0026
Injury Mechanism	0.8387	0.0075
Injury Location	0.9023	0.0048
Time of Injury	0.9974	0.0012

Table 8: Grok-3: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2565	0.0062
Avg Field Accuracy	0.8611	0.0014

Table 9: Cohere-Command-A: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8658	0.0030
Industry	0.9416	0.0039
Severity	0.7026	0.0050
Mode of Transport	0.9445	0.0032
Cause of Injury	0.9968	0.0000
Injury Mechanism	0.6071	0.0085
Injury Location	0.8303	0.0039
Time of Injury	1.0000	0.0000

Table 10: Cohere-Command-A: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4281	0.0186
Avg Field Accuracy	0.8761	0.0107

Table 11: Deepseek R1: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8410	0.0171
Industry	0.9371	0.0044
Severity	0.6913	0.0145
Mode of Transport	0.9516	0.0097
Cause of Injury	0.9571	0.0178
Injury Mechanism	0.8035	0.0210
Injury Location	0.8639	0.0174
Time of Injury	0.9632	0.0158

Table 12: Deepseek R1: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4516	0.0000
Avg Field Accuracy	0.9082	0.0010

Table 13: Claude-Opus-4.5: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9226	0.0026
Industry	0.9731	0.0015
Severity	0.7484	0.0026
Mode of Transport	0.9387	0.0046
Cause of Injury	1.0000	0.0000
Injury Mechanism	0.8022	0.0015
Injury Location	0.8839	0.0026
Time of Injury	0.9968	0.0000

Table 14: Claude-Opus-4.5: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4204	0.0015
Avg Field Accuracy	0.9007	0.0002

Table 15: Claude-Sonnet-4.5: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8968	0.0000
Industry	0.9677	0.0000
Severity	0.7108	0.0015
Mode of Transport	0.9484	0.0000
Cause of Injury	0.9935	0.0000
Injury Mechanism	0.8161	0.0000
Injury Location	0.8946	0.0015
Time of Injury	0.9774	0.0000

Table 16: Claude-Sonnet-4.5: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4763	0.0135
Avg Field Accuracy	0.9151	0.0023

Table 17: Kiwi K2 Thinking: Exact-Match & Average Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8871	0.0026
Industry	0.9452	0.0046
Severity	0.7602	0.0152
Mode of Transport	0.9710	0.0026
Cause of Contact	0.9989	0.0015
Injury Mechanism	0.8570	0.0080
Injury Location	0.9011	0.0040
Time of Injury	1.0000	0.0000

Table 18: Kiwi K2 Thinking: Per-Field Accuracy

On-Premise Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0000	0.0000
Avg Field Accuracy	0.5336	0.0043

Table 19: Mistral-7B-IT-v0.3: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.3148	0.0109
Industry	0.7052	0.0171
Severity	0.4894	0.0134
Mode of Transport	0.6094	0.0107
Cause of Injury	0.9655	0.0046
Injury Mechanism	0.2923	0.0139
Injury Location	0.2339	0.0149
Time of Injury	0.6584	0.0276

Table 20: Mistral-7B-IT-v0.3: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0255	0.0071
Avg Field Accuracy	0.6324	0.0078

Table 21: Llama-3.1-8B-IT: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.3752	0.0161
Industry	0.6900	0.0139
Severity	0.5532	0.0310
Mode of Transport	0.6642	0.0169
Cause of Injury	0.7935	0.0125
Injury Mechanism	0.6000	0.0178
Injury Location	0.4206	0.0148
Time of Injury	0.9623	0.0074

Table 22: Llama-3.1-8B-IT: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0597	0.0030
Avg Field Accuracy	0.7126	0.0024

Table 23: Qwen3-4B-IT-2507: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.3877	0.0094
Industry	0.8784	0.0056
Severity	0.6513	0.0095
Mode of Transport	0.8894	0.0038
Cause of Injury	0.8261	0.0124
Injury Mechanism	0.5619	0.0069
Injury Location	0.5123	0.0061
Time of Injury	0.9935	0.0000

Table 24: Qwen3-4B-IT-2507: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0029	0.0034
Avg Field Accuracy	0.5622	0.0028

Table 25: Granite-3.1-8b-IT: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.3558	0.0123
Industry	0.8226	0.0035
Severity	0.5184	0.0084
Mode of Transport	0.5916	0.0134
Cause of Injury	0.6326	0.0115
Injury Mechanism	0.3045	0.0123
Injury Location	0.3035	0.0098
Time of Injury	0.9687	0.0068

Table 26: Granite-3.1-8b-IT: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.1548	0.0000
Avg Field Accuracy	0.7968	0.0000

Table 27: phi-4: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.7161	0.0000
Industry	0.8774	0.0000
Severity	0.6161	0.0000
Mode of Transport	0.9194	0.0000
Cause of Injury	0.9677	0.0000
Injury Mechanism	0.6323	0.0000
Injury Location	0.6806	0.0000
Time of Injury	0.9645	0.0000

Table 28: phi-4: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0473	0.0030
Avg Field Accuracy	0.6999	0.0019

Table 29: Mistral-3-3B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.5387	0.0026
Industry	0.8376	0.0085
Severity	0.6462	0.0106
Mode of Transport	0.7086	0.0130
Cause of Contact	0.8032	0.0147
Injury Mechanism	0.6538	0.0245
Injury Location	0.4301	0.0110
Time of Injury	0.9806	0.0026

Table 30: Mistral-3-8B: Per-Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2194	0.0095
Avg Field Accuracy	0.8233	0.0033

Table 31: Mistral-3-8B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.7667	0.0080
Industry	0.9043	0.0092
Severity	0.6742	0.0026
Mode of Transport	0.9183	0.0135
Cause of Contact	0.8344	0.0055
Injury Mechanism	0.7581	0.0190
Injury Location	0.7366	0.0130
Time of Injury	0.9935	0.0000

Table 32: Mistral-3-8B: Per-Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2817	0.0119
Avg Field Accuracy	0.8505	0.0022

Table 33: Mistral-3-14B: Exact-Match & Average Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.7817	0.0055
Industry	0.9194	0.0026
Severity	0.7247	0.0124
Mode of Transport	0.9237	0.0066
Cause of Contact	0.8957	0.0110
Injury Mechanism	0.7710	0.0070
Injury Location	0.7989	0.0124
Time of Injury	0.9892	0.0040

Table 34: Mistral-3-14B: Per-Field Accuracy

Fine-Tuned Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3910	0.0204
Avg Field Accuracy	0.8879	0.0043

Table 35: Llama-3.1-8B-IT (Fine-Tuned): Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8952	0.0070
Industry	0.9435	0.0063
Severity	0.6645	0.0151
Mode of Transport	0.9429	0.0038
Cause of Injury	0.9968	0.0029
Injury Mechanism	0.8129	0.0101
Injury Location	0.8513	0.0097
Time of Injury	0.9961	0.0032

Table 36: Llama-3.1-8B-IT (Fine-Tuned): Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2819	0.0179
Avg Field Accuracy	0.8580	0.0039

Table 37: Mistral-7B-IT-v0.3 (Fine-Tuned): Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8719	0.0108
Industry	0.9439	0.0054
Severity	0.5916	0.0198
Mode of Transport	0.8974	0.0069
Cause of Injury	0.9894	0.0035
Injury Mechanism	0.7545	0.0102
Injury Location	0.8177	0.0072
Time of Injury	0.9977	0.0025

Table 38: Mistral-7B-IT-v0.3 (Fine-Tuned): Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3323	0.0282
Avg Field Accuracy	0.8761	0.0113

Table 39: Granite-3.1-8b-IT (Fine-Tuned): Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9006	0.0198
Industry	0.9529	0.0144
Severity	0.6587	0.0271
Mode of Transport	0.9213	0.0041
Cause of Injury	0.9639	0.0055
Injury Mechanism	0.7890	0.0146
Injury Location	0.8223	0.0195
Time of Injury	1.0000	0.0000

Table 40: Granite-3.1-8b-IT (Fine-Tuned): Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2387	0.0184
Avg Field Accuracy	0.8331	0.0035

Table 41: Qwen3-4B-IT-2507 (Fine-Tuned): Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.7739	0.0097
Industry	0.8974	0.0050
Severity	0.6345	0.0084
Mode of Transport	0.9190	0.0030
Cause of Injury	0.9816	0.0052
Injury Mechanism	0.7135	0.0103
Injury Location	0.7484	0.0115
Time of Injury	0.9968	0.0000

Table 42: Qwen3-4B-IT-2507 (Fine-Tuned): Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3957	0.0219
Avg Field Accuracy	0.8883	0.0032

Table 43: Mistral-3-3B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8710	0.0137
Industry	0.9258	0.0079
Severity	0.7215	0.0230
Mode of Transport	0.9355	0.0046
Cause of Injury	0.9882	0.0015
Injury Mechanism	0.8215	0.0055
Injury Location	0.8462	0.0106
Time of Injury	0.9968	0.0000

Table 44: Mistral-3-3B: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4624	0.0100
Avg Field Accuracy	0.9079	0.0033

Table 45: Mistral-3-8B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9108	0.0055
Industry	0.9656	0.0040
Severity	0.6731	0.0145
Mode of Transport	0.9430	0.0055
Cause of Injury	0.9989	0.0015
Injury Mechanism	0.8849	0.0150
Injury Location	0.8903	0.0070
Time of Injury	0.9968	0.0000

Table 46: Mistral-3-8B: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4968	0.0095
Avg Field Accuracy	0.9156	0.0005

Table 47: Mistral-3-14B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9151	0.0080
Industry	0.9613	0.0000
Severity	0.7419	0.0095
Mode of Transport	0.9516	0.0026
Cause of Injury	0.9957	0.0015
Injury Mechanism	0.8699	0.0076
Injury Location	0.8925	0.0030
Time of Injury	0.9968	0.0000

Table 48: Mistral-3-14B: Per Field Accuracy

Few-Shot: Cloud Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.5075	0.0040
Avg Field Accuracy	0.9208	0.0008

Table 49: GPT-5: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.9430	0.0066
Industry	0.9505	0.0015
Severity	0.7505	0.0030
Mode of Transport	0.9828	0.0055
Cause of Injury	0.9978	0.0015
Injury Mechanism	0.8484	0.0046
Injury Location	0.8935	0.0053
Time of Injury	1.0000	0.0000

Table 50: GPT-5: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4430	0.0100
Avg Field Accuracy	0.9001	0.0035

Table 51: GPT-4o: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8796	0.0159
Industry	0.9366	0.0085
Severity	0.7473	0.0080
Mode of Transport	0.9516	0.0070
Cause of Injury	0.9935	0.0026
Injury Mechanism	0.8462	0.0092
Injury Location	0.8462	0.0106
Time of Injury	1.0000	0.0000

Table 52: GPT-4o: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4957	0.0130
Avg Field Accuracy	0.9134	0.0015

Table 53: Grok-3: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.9043	0.0040
Industry	0.9495	0.0030
Severity	0.7624	0.0015
Mode of Transport	0.9613	0.0026
Cause of Injury	1.0000	0.0000
Injury Mechanism	0.8269	0.0076s
Injury Location	0.9032	0.0026
Time of Injury	1.0000	0.0000

Table 54: Grok-3: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3290	0.0026
Avg Field Accuracy	0.8738	0.0003

Table 55: Cohere-Command-A: Overall and Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8774	0.0046
Industry	0.9366	0.0015
Severity	0.7065	0.0026
Mode of Transport	0.9323	0.0026
Cause of Injury	1.0000	0.0000
Injury Mechanism	0.6871	0.0026
Injury Location	0.8505	0.0015
Time of Injury	1.0000	0.0000

Table 56: Cohere-Command-A: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4710	0.0121
Avg Field Accuracy	0.8962	0.0082

Table 57: Deepseek R1: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8699	0.0085
Industry	0.9161	0.0121
Severity	0.7237	0.0085
Mode of Transport	0.9495	0.0076
Cause of Injury	0.9849	0.0066
Injury Mechanism	0.8376	0.0066
Injury Location	0.9011	0.0150
Time of Injury	0.9871	0.0091

Table 58: Deepseek R1: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4935	0.0115
Avg Field Accuracy	0.9129	0.0029

Table 59: Gemini flash 2.5: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8989	0.0055
Industry	0.9677	0.0026
Severity	0.7430	0.0040
Mode of Transport	0.9473	0.0040
Cause of Injury	0.9978	0.0015
Injury Mechanism	0.8591	0.0119
Injury Location	0.8957	0.0040
Time of Injury	0.9935	0.0000

Table 60: Gemini flash 2.5: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4333	0.0015
Avg Field Accuracy	0.9036	0.0003

Table 61: Claude-Sonnet-4.5: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8839	0.0000
Industry	0.9613	0.0000
Severity	0.7323	0.0000
Mode of Transport	0.9452	0.0026
Cause of Injury	0.9968	0.0000
Injury Mechanism	0.8161	0.0000
Injury Location	0.9065	0.0000
Time of Injury	0.9871	0.0000

Table 62: Claude-Sonnet-4.5: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4516	0.0000
Avg Field Accuracy	0.9082	0.0010

Table 63: Claude-Opus-4.5: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.9226	0.0026
Industry	0.9731	0.0015
Severity	0.7484	0.0026
Mode of Transport	0.9387	0.0046
Cause of Injury	1.0000	0.0000
Injury Mechanism	0.8022	0.0015
Injury Location	0.8839	0.0026
Time of Injury	0.9968	0.0000

Table 64: Claude-Opus-4.5: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.5081	0.0016
Avg Field Accuracy	0.9212	0.0006

Table 65: Kiwi K2 High Accuracy: Exact-Match & Average Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9387	0.0032
Industry	0.9500	0.0048
Severity	0.7565	0.0016
Mode of Transport	0.9839	0.0032
Cause of Contact	1.0000	0.0000
Injury Mechanism	0.8435	0.0016
Injury Location	0.8968	0.0000
Time of Injury	1.0000	0.0000

Table 66: Kiwi K2 High Accuracy: Per-Field Accuracy

Few-Shot: On-Premise Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0419	0.0026
Avg Field Accuracy	0.6827	0.0043

Table 67: Mistral-7B-IT-v0.3: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.6086	0.0106
Industry	0.7430	0.0092
Severity	0.3882	0.0015
Mode of Transport	0.8333	0.0085
Cause of Injury	0.9323	0.0070
Injury Mechanism	0.5602	0.0171
Injury Location	0.4280	0.0175
Time of Injury	0.9677	0.0046

Table 68: Mistral-7B-IT-v0.3: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0978	0.0040
Avg Field Accuracy	0.7492	0.0054

Table 69: Llama-3.1-8B-IT: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.6774	0.0121
Industry	0.8602	0.0119
Severity	0.5806	0.0199
Mode of Transport	0.8473	0.0040
Cause of Injury	0.8957	0.0239
Injury Mechanism	0.6247	0.0145
Injury Location	0.5495	0.0250
Time of Injury	0.9581	0.0070

Table 70: Llama-3.1-8B-IT: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.0785	0.0080
Avg Field Accuracy	0.7478	0.0031

Table 71: Granite-3.1-8b-IT: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.7075	0.0061
Industry	0.8753	0.0030
Severity	0.5516	0.0158
Mode of Transport	0.8613	0.0091
Cause of Injury	0.9613	0.0046
Injury Mechanism	0.5570	0.0154
Injury Location	0.4903	0.0095
Time of Injury	0.9785	0.0040

Table 72: Granite-3.1-8b-IT: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.1151	0.0040
Avg Field Accuracy	0.7688	0.0008

Table 73: Qwen3-4B-IT-2507: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.6505	0.0030
Industry	0.8946	0.0015
Severity	0.6269	0.0040
Mode of Transport	0.8989	0.0015
Cause of Injury	0.9140	0.0040
Injury Mechanism	0.6538	0.0080
Injury Location	0.5409	0.0061
Time of Injury	0.9710	0.0000

Table 74: Qwen3-4B-IT-2507: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2903	0.0000
Avg Field Accuracy	0.8516	0.0000

Table 75: phi-4: Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8194	0.0000
Industry	0.9258	0.0000
Severity	0.6742	0.0000
Mode of Transport	0.9161	0.0000
Cause of Injury	0.9935	0.0000
Injury Mechanism	0.7226	0.0000
Injury Location	0.7645	0.0000
Time of Injury	0.9968	0.0000

Table 76: phi-4: Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3312	0.0179
Avg Field Accuracy	0.8675	0.0026

Table 77: Mistral-3-14B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8086	0.0055
Industry	0.9215	0.0110
Severity	0.7323	0.0095
Mode of Transport	0.9258	0.0121
Cause of Contact	0.9828	0.0015
Injury Mechanism	0.7806	0.0070
Injury Location	0.7989	0.0130
Time of Injury	0.9892	0.0015

Table 78: Mistral-3-14B: Per-Field Accuracy

Metric / Field	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.1398	0.0085
Avg Field Accuracy	0.7737	0.0058

Table 79: Mistral-3-3B: Exact-Match & Avg. Field Accuracy

Metric / Field	Average (Mean)	Standard Deviation
Activity	0.6688	0.0133
Industry	0.8323	0.0079
Severity	0.6860	0.0110
Mode of Transport	0.8323	0.0091
Cause of Contact	0.9462	0.0055
Injury Mechanism	0.6817	0.0159
Injury Location	0.5495	0.0085
Time of Injury	0.9925	0.0015

Table 80: Mistral-3-3B: Per-Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2634	0.0199
Avg Field Accuracy	0.8491	0.0036

Table 81: Mistral-3-8B: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8194	0.0026
Industry	0.9000	0.0053
Severity	0.6591	0.0150
Mode of Transport	0.9226	0.0026
Cause of Contact	0.9527	0.0055
Injury Mechanism	0.7495	0.0080
Injury Location	0.7946	0.0055
Time of Injury	0.9946	0.0015

Table 82: Mistral-3-8B: Per-Field Accuracy

Few-Shot: Fine-Tuned Models

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2602	0.0145
Avg Field Accuracy	0.8469	0.0027

Table 83: Mistral-7B-IT-v0.3 (Fine-Tuned): Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8538	0.0130
Industry	0.9204	0.0040
Severity	0.5462	0.0201
Mode of Transport	0.9032	0.0026
Cause of Injury	0.9892	0.0030
Injury Mechanism	0.7591	0.0085
Injury Location	0.8032	0.0026
Time of Injury	1.0000	0.0000

Table 84: Mistral-7B-IT-v0.3 (Fine-Tuned): Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3269	0.0106
Avg Field Accuracy	0.8765	0.0019

Table 85: Llama-3.1-8B-IT (Fine-Tuned): Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8774	0.0095
Industry	0.9290	0.0053
Severity	0.6656	0.0100
Mode of Transport	0.9290	0.0026
Cause of Injury	0.9935	0.0026
Injury Mechanism	0.7935	0.0091
Injury Location	0.8333	0.0066
Time of Injury	0.9903	0.0000

Table 86: Llama-3.1-8B-IT (Fine-Tuned): Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3097	0.0173
Avg Field Accuracy	0.8749	0.0016

Table 87: Granite-3.1-8b-IT (Fine-Tuned): Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.8785	0.0030
Industry	0.9484	0.0046
Severity	0.6817	0.0110
Mode of Transport	0.9290	0.0000
Cause of Injury	0.9785	0.0015
Injury Mechanism	0.7688	0.0055
Injury Location	0.8140	0.0030
Time of Injury	1.0000	0.0000

Table 88: Granite-3.1-8b-IT (Fine-Tuned): Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.2161	0.0070
Avg Field Accuracy	0.8220	0.0017

Table 89: Qwen3-4B-IT-2507 (Fine-Tuned): Exact-Match & Avg. Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Activity	0.7602	0.0055
Industry	0.8989	0.0055
Severity	0.6269	0.0100
Mode of Transport	0.8968	0.0026
Cause of Injury	0.9720	0.0040
Injury Mechanism	0.6935	0.0079
Injury Location	0.7409	0.0015
Time of Injury	0.9871	0.0000

Table 90: Qwen3-4B-IT-2507 (Fine-Tuned): Per Field Accuracy (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.3613	0.0079
Avg Field Accuracy	0.8716	0.0013

Table 91: Mistral-3-3B Few-Shot: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.8419	0.0091
Industry	0.9097	0.0095
Severity	0.7032	0.0053
Mode of Transport	0.9376	0.0030
Cause of Injury	0.9828	0.0015
Injury Mechanism	0.7935	0.0046
Injury Location	0.8097	0.0105
Time of Injury	0.9946	0.0030

Table 92: Mistral-3-3B Few-Shot: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4129	0.0026
Avg Field Accuracy	0.8964	0.0018

Table 93: Mistral-3-8B Few-Shot: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9043	0.0055
Industry	0.9215	0.0040
Severity	0.6828	0.0106
Mode of Transport	0.9505	0.0061
Cause of Injury	0.9957	0.0015
Injury Mechanism	0.8538	0.0030
Injury Location	0.8677	0.0046
Time of Injury	0.9946	0.0015

Table 94: Mistral-3-8B Few-Shot: Per Field Accuracy

Metric	Average (Mean)	Standard Deviation
Exact-Match Accuracy	0.4957	0.0066
Avg Field Accuracy	0.9151	0.0013

Table 95: Mistral-3-14B Few-Shot: Exact-Match & Avg. Field Accuracy

Metric	Average (Mean)	Standard Deviation
Activity	0.9118	0.0066
Industry	0.9581	0.0053
Severity	0.7430	0.0119
Mode of Transport	0.9559	0.0061
Cause of Injury	1.0000	0.0000
Injury Mechanism	0.8667	0.0061
Injury Location	0.8882	0.0040
Time of Injury	0.9968	0.0000

Table 96: Mistral-3-14B Few-Shot: Per Field Accuracy

Performance Tables

Cloud Models

Metric	Average (Mean)	Standard Deviation
Mean	5.9917	0.0487
Median	1.9032	0.2703
Standard Deviation	11.8844	0.8683
Min	1.4068	0.1152
Max	45.9160	1.8426

Table 97: GPT-5: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0	0.0
Mean	5.9576	0.0221
Median	1.6532	0.1208
Standard Deviation	12.5684	0.6797
Min	1.2681	0.0711
Max	46.7635	1.7512

Table 98: GPT-4o: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean	9.5510	2.6226
Median	8.8826	2.1114
Standard Deviation	3.9456	1.6165
Min	3.6741	0.5730
Max	29.4751	12.6160

Table 99: Gemini flash 2.5: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	6.3161	1.3602
Median (sec/case)	5.2288	0.1892
Std Dev	3.7391	3.3536
Min	3.1709	0.0625
Max	34.1928	19.4127

Table 100: Grok-3: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	5.9665	0.4888
Median	5.7293	0.0744
Standard Deviation	1.3843	2.8331
Min	4.7280	0.0983
Max	23.4334	47.8005

Table 101: Cohere-Command-A: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean	21.4417	10.9532
Median	15.9079	6.4087
Std Dev	16.4850	11.5128
Min	6.5855	0.7437
Max	118.5725	75.3603

Table 102: Deepseek-R1: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0000	0.0000
Mean	5.5729	0.0360
Median	5.4632	0.0518
Standard Deviation	0.5646	0.0598
Min	4.6003	0.0125
Max	8.9094	1.6805

Table 103: Claude-Opus-4.5:Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0000	0.0000
Mean	5.8027	0.0362
Median	5.6896	0.0515
Standard Deviation	0.6091	0.0145
Min	4.6965	0.0583
Max	10.8797	0.8597

Table 104: Claude-Sonnet-4.5: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	5.9584	0.0050
Median time per case	1.8346	0.0120
Standard Deviation	12.2676	0.1657
Min	1.5593	0.0252
Max	45.2918	0.2033

Table 105: Kiwi K2 Thinking: Inference Time Metrics (sec / case)

On-Premise Models

Metric	Average (Mean)	Standard Deviation
Mean	7.7073	5.3377
Median	5.8095	0.0117
Std Dev	31.9940	99.8586
Min	5.0054	0.0537
Max	563.2453	1759.3577

Table 106: Mistral-7B-IT-v0.3: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	5.4925	0.0135
Median time per case	5.3682	0.0132
Standard Deviation	0.5376	0.0198
Min	4.3342	0.1720
Max	6.9730	0.0843

Table 107: Llama-3.1-8B-IT: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	3.9874	0.0108
Median time per case	3.9227	0.0226
Standard Deviation	0.4073	0.0079
Min	3.0970	0.1052
Max	5.2492	0.1652

Table 108: Qwen3-4B-IT-2507: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	6.9197	0.0195
Median time per case	6.7648	0.0163
Standard Deviation	0.7041	0.0330
Min	5.4182	0.1923
Max	9.4768	0.7258

Table 109: Granite-3.1-8b-IT: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	10.7790	0.0006
Median time per case	10.7644	0.0122
Standard Deviation	0.6249	0.0003
Min	9.2034	0.0073
Max	12.8890	0.0082

Table 110: phi-4: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	10.4049	0.0131
Median (sec/case)	10.3112	0.0137
Std Dev	0.6229	0.0077
Min	8.8063	0.0275
Max	12.4249	0.0836

Table 111: Mistral-3-3B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	6.8492	0.0148
Median (sec/case)	6.7948	0.0127
Std Dev	0.4612	0.0029
Min	5.6658	0.0076
Max	8.3351	0.0199

Table 112: Mistral-3-8B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	9.5756	0.0155
Median time per case	9.5091	0.0166
Standard Deviation	0.5512	0.0180
Min	8.0049	0.0824
Max	11.3153	0.1744

Table 113: Mistral-3-14B: Inference Time Metrics (sec / case)

Fine-Tuned Models

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	9.8781	0.0315
Median (sec/case)	9.5854	0.0572
Standard Deviation	1.0939	0.0242
Min	7.9447	0.1052
Max	13.1757	0.7173

Table 114: Granite-3.1-8b-IT (Fine-Tuned): Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	6.9479	0.0073
Median (sec/case)	6.8306	0.0052
Standard Deviation	0.6362	0.0140
Min	5.6156	0.0302
Max	9.1409	0.0063

Table 115: Llama-3.1-8B-IT (Fine-Tuned): Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	8.4170	0.0476
Median (sec/case)	8.2428	0.0586
Std Dev	0.9616	0.1462
Min	6.7001	0.0615
Max	14.7934	3.9779

Table 116: Mistral-7B-IT-v0.3 (Fine-Tuned): Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean	5.9640	0.0071
Median	5.8715	0.0071
Std Dev	0.6027	0.0076
Min	4.7287	0.0142
Max	7.7495	0.0149

Table 117: Qwen3-4B-IT-2507: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	10.4049	0.0131
Median (sec/case)	10.3112	0.0137
Std Dev	0.6229	0.0077
Min	8.8063	0.0275
Max	12.4249	0.0836

Table 118: Mistral-3-3B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	6.8492	0.0148
Median (sec/case)	6.7948	0.0127
Std Dev	0.4612	0.0029
Min	5.6658	0.0076
Max	8.3351	0.0199

Table 119: Mistral-3-8B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	9.9181	0.0045
Median (sec/case)	9.8307	0.0132
Std Dev	0.6501	0.0095
Min	8.2401	0.0140
Max	12.0366	0.1444

Table 120: Mistral-3-14B: Inference Time Metrics

Few-Shot: Cloud Models

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	7.5392	0.0408
Median (sec/case)	2.2037	0.0760
Standard Deviation	13.2166	0.6108
Min	1.6415	0.1373
Max	53.2134	12.7359

Table 121: GPT-5: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0000	0.0000
Mean (sec/case)	7.5468	0.1198
Median (sec/case)	1.5757	0.2427
Standard Deviation	14.3238	2.4492
Min	1.1826	0.0250
Max	53.9183	4.4103

Table 122: GPT-4o: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	11.1309	0.0000
Median (sec/case)	2.7429	0.0000
Standard Deviation	17.5759	0.0000
Min	1.7241	0.0000
Max	55.8235	0.0000

Table 123: Claude-Haiku-4.5: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	7.2528	0.3865
Median	6.8184	0.3722
Std Dev	2.7257	0.2490
Min	3.3708	0.2530
Max	26.3480	8.0221

Table 124: Gemini flash 2.5: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	6.3161	1.3602
Median (sec/case)	5.2288	0.1892
Std Dev	3.7391	3.3536
Min	3.1709	0.0625
Max	34.1928	19.4127

Table 125: Grok-3: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0000	0.0000
Mean	5.5729	0.0360
Median	5.4632	0.0518
Standard Deviation	0.5646	0.0598
Min	4.6003	0.0125
Max	8.9094	1.6805

Table 126: Claude-Opus-4.5:Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
N (Count)	310.0000	0.0000
Mean	4.9310	0.0556
Median	4.7791	0.0583
Standard Deviation	0.7286	0.0673
Min	3.8195	0.1167
Max	8.4996	1.4602

Table 127: Claude-Sonnet-4.5: Inference Time Metrics (Few-Shot)

Few-Shot: On-Premise Models

Metric	Average (Mean)	Standard Deviation
Mean	10.5470	0.0164
Median	10.3597	0.0410
Std Dev	0.8839	0.0155
Min	8.8858	0.0034
Max	13.3081	0.2592

Table 128: Granite-3.1-8b-IT: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	7.7111	0.0124
Median	7.5281	0.0105
Std Dev	0.7043	0.0139
Min	6.3793	0.1974
Max	10.0501	0.0094

Table 129: Llama-3.1-8B-IT: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	9.2306	0.0097
Median	9.1659	0.0090
Std Dev	0.7668	0.0347
Min	5.6995	0.3247
Max	11.8207	0.1753

Table 130: Mistral-7B-IT-v0.3: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	6.2595	0.0139
Median	6.1526	0.0128
Std Dev	0.6809	0.0110
Min	4.7596	0.0201
Max	8.4799	0.0232

Table 131: Qwen3-4B-IT-2507: Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean time per case	10.1214	0.0378
Median time per case	10.0446	0.0370
Standard Deviation	0.6999	0.2507
Min	8.4401	0.3141
Max	17.2064	6.4770

Table 132: Mistral-3-14B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean (sec/case)	10.4049	0.0131
Median (sec/case)	10.3112	0.0137
Std Dev	0.6229	0.0077
Min	8.8063	0.0275
Max	12.4249	0.0836

Table 133: Mistral-3-3B: Inference Time Metrics

Metric	Average (Mean)	Standard Deviation
Mean time per case	6.9519	0.0108
Median time per case	6.8897	0.0107
Standard Deviation	0.3927	0.0091
Min	5.7565	0.2144
Max	8.3157	0.1208

Table 134: Mistral-3-8B: Inference Time Metrics

Few-Shot: Fine-Tuned Models

Metric	Average (Mean)	Standard Deviation
Mean	11.1647	0.0009
Median	10.8178	0.0272
Std Dev	1.2186	0.0261
Min	9.2580	0.2027
Max	15.1534	1.2638

Table 135: Granite-3.1-8b-IT (Fine-Tuned): Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	7.8614	0.0127
Median	7.7197	0.0143
Std Dev	0.6888	0.0032
Min	6.3273	0.0065
Max	10.0588	0.0724

Table 136: Llama-3.1-8B-IT (Fine-Tuned): Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	9.4784	0.0115
Median	9.2973	0.0133
Std Dev	0.9494	0.0137
Min	7.2660	0.4772
Max	12.3273	0.0038

Table 137: Mistral-7B-IT-v0.3 (Fine-Tuned): Inference Time Metrics (Few-Shot)

Metric	Average (Mean)	Standard Deviation
Mean	6.9516	0.0061
Median	6.8036	0.0012
Std Dev	0.7174	0.0045
Min	5.3235	0.0684
Max	9.3949	0.0161

Table 138: Qwen3-4B-IT-2507 (Fine-Tuned): Inference Time Metrics (Few-Shot)